T-Mobile Customer Selective Roaming Agreement

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Version 1.2

Updated 8/23/2024
Submitted: April 2024TBD

T-Mobile Confidential

Customer Selective Roaming Agreement - Concept Idea

Allow a customer to choose their roaming partner and agreement, costs, quality, and policy.

- **Problem Statement**: Today, carriers provide a 'default' roaming plan to customers when they go out of in-carrier service. This forces the customer into a roaming agreement and costs that they did not choose or have a choice. Their only option is to turn off roaming. This IP provides a customer selection and roaming agreement technology to allow a customer to obtain, review, select, and approve a carrier's alternative roaming partner choices.
  - Solves customer roaming costs and ensures direct choice and notification of all costs.
  - Resolves US border to non-US roaming agreements.
  - Resolves private-party roaming agreements for costs (e.g. Cruise Ships vs. land tower roaming).
  - Resolves European high-frequency roaming conditions to stay 'static' with one preferred carrier roaming partner.

- **Use Cases**
- As a T-Mobile customer with an activated device, when I go out of my own carrier's service capability, I want to see a choice of all roaming partners that may offer a roaming service.
- As a T-Mobile customer with an activated device, I want to be able to review the service, quality, plan, consumer rating, and costs to allow me to decide the best roaming plan to use while I am in a roaming situation.
- T-Mobile as a provider needs to be able to offer customers more choices and greater satisfaction for roaming partners, and also ensure our own roaming capabilities back to other provides can be selected by external non-T-Mobile customers for preferred roaming.

Technical Approach

- T-Mobile roaming agreements shall be presented using an on-device client (agent) that would aggregate all carrier localized agreements within their region of travel.
- The agreement selection shall be presented as a list of providers with supporting metrics for decision-making. (E.g. quality of service, customer ratings, compatibility, costs, coverage, etc.)
- Upon device startup, a roaming proxy agent shall be preloaded onto the device that will manage and handle the customer experience and roaming partner configurations. This needs to be dynamically updated as the customer travels. This roaming configuration shall be updated frequently by the backend T-Mobile servers, domestic and international (e.g. Deutche Telekom).
- Upon a customer selective roaming agreement, the device shall auto-reprogram to the new configuration and set roaming rates accordingly for the customer with T-Mobile.

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High Level User Experience

On-Device Client Roaming Agent

- On-Device application to manage user experiences and CSRA functions.
- App handles carrier billing / payment processing.
- App manages synchronization of Roaming Partner Server "Offers" based on geozones.
- App manages all PRL/firmware/client software/authentication to support CSRA. App auto-updates PRLs/radio resets based on completion of offer.
- Roaming choices are presented as a menu offering (offer/bids from external carriers).
- After customer choice is made, device auto-programs to new targeted roaming partner configuration.

Sends real-time roaming agreement selection choices per customer travels.

T-Mobile Roaming Partner Server

T-Mobile backend continuously uses AI to determine proper roaming configurations and customer preferent selection options.

Roaming "Aggregator"

- Roaming Meta Data → Verizon Roaming Plans
- Roaming Meta Data → AT&T Roaming Plans
- Roaming Meta Data → Private Party Roaming Plans
- Roaming Meta Data → International Roaming Plans

CSRA – Client Approach

- T-Mobile creates an on-device client app for CSRA management and PRL updating (including radio reset capabilities) and preloads on every T-Mobile/Metro engineered device.
- T-Mobile roaming agreements shall be presented using an on-device client (agent) that would aggregate all carrier localized agreements within their region of travel. This client presents these choices as a 'Roaming Offer' to customers to provide best roaming options for a customer out of their normal travel zones.
- The agreement selection shall be presented as a list of providers with supporting metrics for decision-making. (E.g. quality of service, customer ratings, compatibility, costs, coverage, etc.)
- Upon device startup, a roaming proxy agent shall be preloaded onto the device that will manage and handle the customer experience and roaming partner configurations. The client shall be dynamically updated as the customer travels. This roaming configuration shall be updated frequently by the backend T-Mobile servers, domestic and international (e.g. Deutche Telekom). The client shall perform periodic updates based on their new Lat/Log location capabilities to maintain accuracy of the CSRA configurations. This includes offshore locations (such as traveling on airlines or cruise ships).
- Upon a customer selective roaming agreement, the device shall auto-reprogram to the new configuration and set roaming rates accordingly for the customer with T-Mobile. The client shall process a payment flow with the customer (either carrier billing or immediate payment processing), then proceed to align the device configuration and PRL updates to the device for the roaming configuration. The client shall then reset necessary device configuration files (including updating the device itself), reconfigure necessary firmware/client software, necessary PRL files and configurations, and reset the radios if required. In the final step, the client (agent) shall have an appropriate customer experience to advise the customer of the changes.
- If the travel path of the customer continues outside the defined CSRA PRL zone, the on-device client shall again re-engage with the customer for a new determinate decisions to re-choose their CSRA profile. This is an automated on-device Lat/Long, PRL (AI logic) continuous function.

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Client Functions

On-Device Client Roaming APK

Sends real-time roaming configurations between server and client.

T-Mobile Roaming Partner Server

Provides a UX interface to consumer based on roaming conditions.

Preload in firmware of T-Mobile/Metro devices.

On-Device CSRA client reconfigures PRL files, radio, firmware, radio reset

Roaming choices are presented and recorded to consumer account for proper billing.

After customer choice is made, device auto-programs to new targeted roaming partner configuration.

Continuous Lat/Long monitoring for applied CSRA zone. React to out-of-zone changes for new offer.

Meta data is downloaded to client for potential roaming configuration use (periodically).

Handles customer payment processing (carrier biller/CC payment)

Today:
The PRL files are configured at OOBE startup and are not continuously updated. Additionally, customers have no choice on which PRL configuration they can use, thus costs are fixed by the carrier or a predefined carrier-to-carrier pre-arranged agreement.

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Server Functions

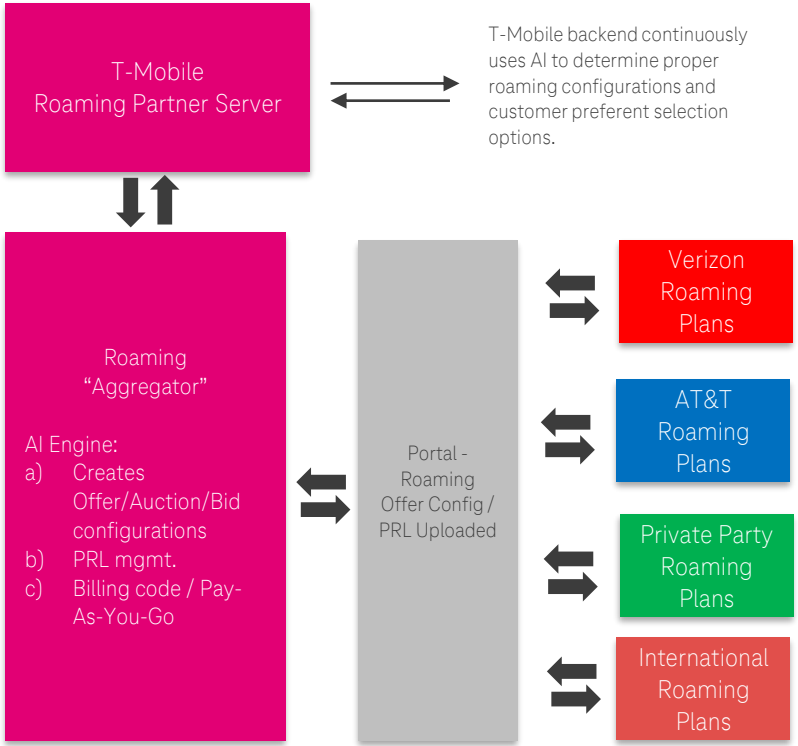
T-Mobile CSRA Server Functions

1) A dynamically configurable, in near-real-time, server-side Offer/Bid management for any North American carrier.

a) Provides a portal/endpoint for carriers to provide auction data as an "offer" selection for on-device client.
b) Carriers create an account and enter their roaming agreement pricing/offer/service levels/band/authentication method, etc.
c) Carriers upload their offer/bid PRL list to T-Mobile.
d) AI logic of T-Mobile servers compiles this geo-area config and offer using multiple region criteria (Lat/Long, zip code, country code/state/territory, etc.)
e) T-Mobile server assigns a biller agreement code (or one-time fee for customer billing). Optionally, a pay-as-you-go payment popup window could also be provided on the client.
f) Server prepares a CSRA meta-data configuration package for on-device roaming client to offer the customer.

Today's Configuration

Roaming agreements today are static, and pre-arranged between carriers as master roaming agreements. Customers have no choice or selectivity to choose (or reduce their overall expenses for roaming) sync the pre-existing agreements are only managed by T-Mobile as a pre-defined configuration.



- PRL config
- GeoZone covered
- Pricing
- Service Levels

CSRA – Backend Approach

- T-Mobile roaming agreements shall be presented using an on-device client (agent) that would aggregate all carrier localized agreements within their region of travel.
- The agreement selection shall be presented as a list of providers with supporting metrics for decision-making. (E.g. quality of service, customer ratings, payment options (billed/immediate payment), compatibility, costs, coverage, etc.)
- Upon a customer selective roaming agreement, the device shall auto-reprogram to the new configuration and set roaming rates accordingly for the customer with T-Mobile.

Inventor Details

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